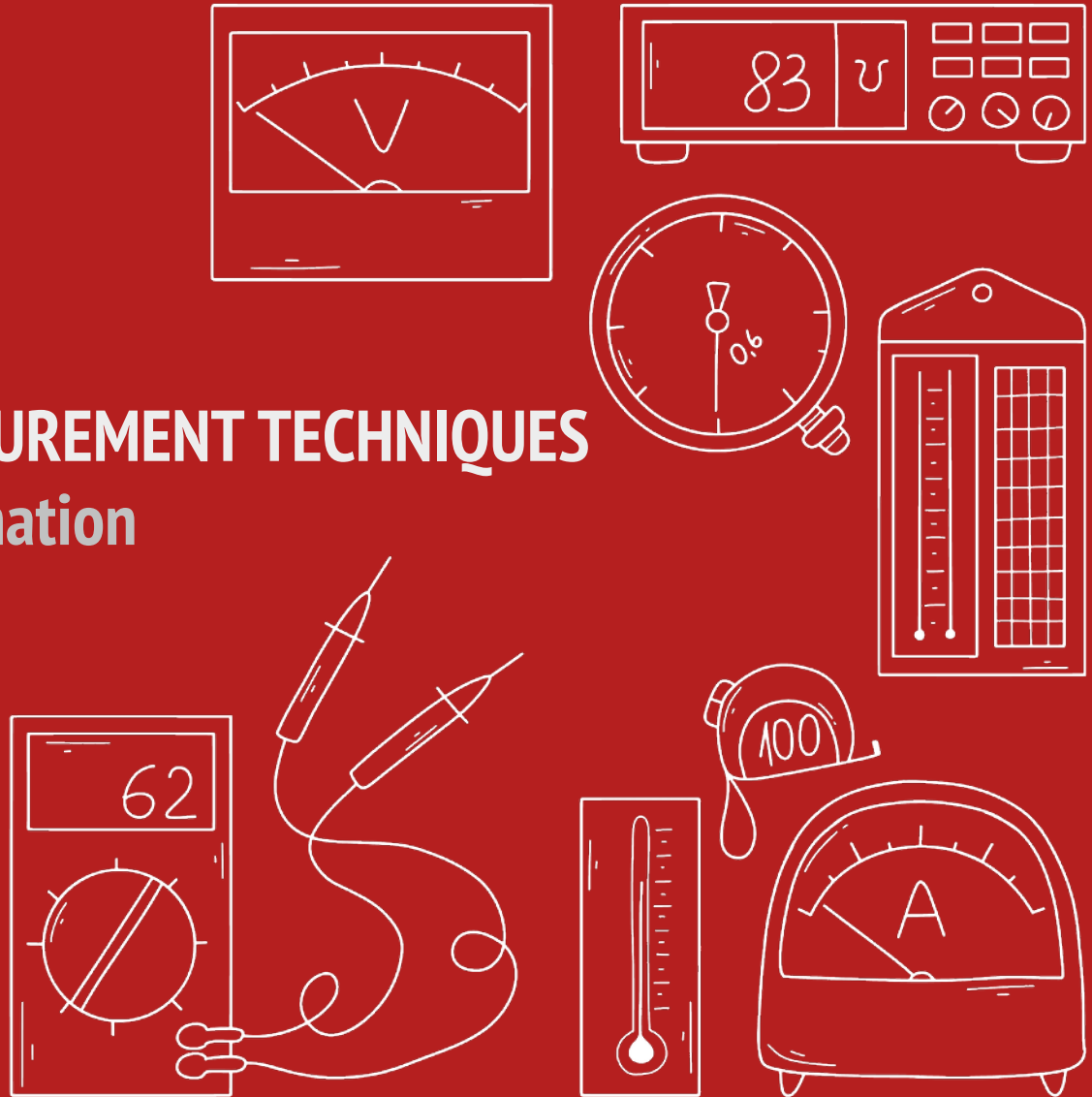


ME-301 MEASUREMENT TECHNIQUES

General information



General information

Course description

Theoretical part

Practical part

Questions

ME-301 Measurement techniques

Nowadays, data science and data-driven methods are widely used in many aspects of our personal and professional lives. Yet, the success of many of these methods is limited by the quality of the data that they are driven by. As future engineers, it is essential not only to learn how to apply, implement, and develop data-driven methods, but also to understand how data is generated, how to accurately measure key quantities, the working principles and limitations of commonly used measurement techniques, the physical limitations of measurement systems, potential sources of error and uncertainty that may arise, and how to debug a measurement system.

The goal of ME301 is to provide you with **theoretical** understanding and **practical** experience of:

- experimental methodology and practice i.e. how to do experiments
- sources of error and uncertainty in acquiring data
- working principles behind common experimental systems for the measurement of physical quantities
- signal processing tools for conditioning and analysing measurement data

ME-301 Measurement techniques

Course summary

This is a  **theoretical** and  **practical** course on basic measurement principles and experimental techniques for observation and measurement of physical variables such as force, strain, temperature, flow velocity, structural deformation and vibrations, etc.

Credits: 4 ECTS

4 ECTS correspond to a total expected workload of $4 \times 30 \text{ h} = 120 \text{ h}$ or roughly 8.5 h/week for 14 weeks. We counted roughly a total estimated workload *per person* in the group of 80 h or a little over 6 h/Friday that we are in session this semester for the practical part and 40 h for the theoretical part.

Assessment methods:

✓ **Final written exam (30 %)**

✓ **Technical report (70 %)**

Course information:



Theoretical part

CONTENT The theoretical part consists of the following chapters:

1. General measurement principles
2. Signal processing
3. Specialised measurement systems
 - a) Flow measurements
 - b) Force, torque, and strain measurements
 - c) Motion and vibration measurements
 - d) Optical absorption measurements and photoelasticity
 - e) Temperature and thermal property measurements

There will be no in-person lectures, except for a master class on *Signal processing* by Dr Müllhaupt. The schedule is on the ME301 Notion website 

LECTURE MATERIAL All course material is presented in form of video lectures by various lecturers that can be found on  Mediaspace.

ASSESSMENT A written exam with multiple-choice questions on the theoretical course material will count for 30 % of your final grade.

Practical part

CONTENT You will learn how to use different measurement techniques and conduct an engineering experiment through a semester long practical group project.

ORGANISATION The practical session take place in the SPOT  on Fridays

Due to the large number of groups, you have been divided into three sessions:

Session A	08:15 - 11:00	Groups 01 to 20
Session B	11:15 - 14:00	Groups 21 to 40
Session C	14:15 - 17:00	Groups 41 to 60

Due to the large number of students, we cannot accommodate any requests for changes.

When all group members from a group in a different session agree with your entire group to switch sessions, you can switch with them informally. **We can not formally accommodate any request for changes.**

If any issues arise, we will insist on the assigned session division.

Room **MED 2 2423** is freely available to you unless it is used for meetings that are indicated in the schedule on . A team of teachers & assistants will be available at the SPOT only.

Practical part

REQUIREMENTS You and your group will:

- select a (simple) experiment inspired by a mechanical engineering problem or physical event among the examples listed



 Rubber band thermodynamics



 Jet flows



 Flapping of flags



 Vortex induced vibrations



 Bouncing of a ball



 Surface gravity waves



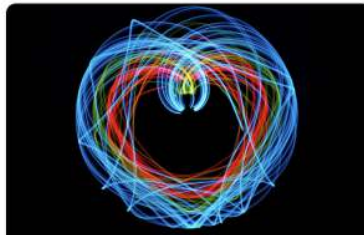
 Oscillation and resonance



 Forces on a sail



 Convection



 Pendulum motion



 Plants interacting with wind



Plants interacting with wind

+ Add a property

Comments

 Add a comment...

Inspiration

Experimental setup

Option 1: in air

Option 2: in water

Experimental systems

A. Image-processing based measure...

B. Sensor measurements

C. Optical field measurement

 The descriptions below are suggestions to get you started. These are not recipes that you just need to follow step-by-step, you will probably have to adapt and modify the set-up and measurements systems to get the best results. Do not be afraid to try different options, before settling on your final configuration. If you have others idea that are not listed here, feel free to pursue them and use your creativity.

Inspiration

Grassy plants waving in the morning breeze, kelp forests swaying in the underwater currents, are all cases of vegetation interacting with natural flows. With this setup we will mimic the behaviour of a blade of grass interacting with wind.



+ :: Experimental setup



Practical part

REQUIREMENTS You and your group will:

- select a (simple) experiment inspired by a mechanical engineering problem or physical event among the examples listed
- build the experimental set-up

Experimental setup

For this experiment, we will induce the deformation of a thin cantilevered strip using the aerodynamic forces generated by flowing air or water.

When the strip is exposed to the flow, it deforms as a consequence of the aerodynamic forces. The strip reconfigures to a curved shape that reduces its drag compared to the standing straight configuration, and you can measure the shape, deformation, forces on the strip, etc.

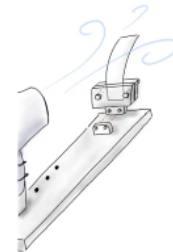
Option 1: in air

General idea

We want to fix the blade so that it protrudes into the flow, with its surface normal to the flow direction. To this end, it can e.g. be clamped between two plates, which will be in turn fastened to your experimental assembly. Use two fixation points at all times to allow repeatable alignment and orientation of the strip.

 All sensors and measurement tools, including load measuring devices, have been intentionally omitted from the conceptual drawing. These must be implemented independently by each group as part of the course.

As a starting point, we suggest the strip to be around a centimetre wide, and protrude around ten centimetres from the base.



Rough sketch of a possible set-up.

Main challenges

- To conduct repeatable measurements:
 - your set-up should be constructed such that all elements can be placed at the same location and orientation from one week to another
 - you should be able to place fix the hairdryer at different fixed distances with respect to the strip
- In real life, we would adapt the measurement systems to the set-up. Due constraints on the availability and costs of equipment, you will have to adapt the set-up to the available measurement systems in this course.

Here, you want to measure a displacement of the strip. If the strip is too stiff, it will not bend and you should use different strips. If you want to measure the drag force, but the signal is too small, you can use a larger strip, or increase the fluid dynamic forces

Parameters that can be varied

- Material of the strip.

 Do not 3D print flat surfaces, strips can be cut from different sheets of plastic or other materials, look through the reusable materials at SPOT, or ask the teaching team.

- Dimensions of the strip (length, width, thickness).
- Shape of the strip (rectangle, tapered).
- Distance from the hairdryer.

Option 2: in water

Practical part

REQUIREMENTS You and your group will:

- select a (simple) experiment inspired by a mechanical engineering problem or physical event among the examples listed
- build the experimental set-up
- conduct measurements

Practical part

REQUIREMENTS You and your group will:

- select a (simple) experiment inspired by a mechanical engineering problem or physical event among the examples listed
- build the experimental set-up
- conduct measurements

■ 1 image-processing based measurement

Image-processing can be part of the evaluation procedure of optical field measurement techniques, but it can also be used to:

- track objects or markers across frames to quantify their **position, displacement, or velocity**
- identifying geometric features (edges, contours) and evaluate the **shape, dimensions, or deformation** of objects

Experimental systems

Here are some suggestions of measurements that you can conduct with this set-up that fit the  Practical part - Requirements of the course. Remember that these are suggestions and you will have to adapt them to the specific dimensions of your set-up. You are also free to select other measurements as long as they fit the requirements.

A. Image-processing based measurement

Example measurements using images and image processing in this type of set-up are:

- optically tracking the edge of the strip in an image sequence
- optically extracting the shape of the deformed strip

Main challenges

- We want to have clear contrast between the feature of interest and the background. In this case, the edge of the strip. Experiment with the positioning of light sources to exploit the strip's transparency and internal reflections in order to light up its edge.
- Use your smartphone camera to obtain a high-speed video of the strip, extract and track the strip's edge and tip position through image processing techniques.
- To increase the resolution of your measurement, the field of view should zoom in as much as possible on the region where interesting things happen.
- In order to obtain sharp images despite of fast movement, we need to record images with a short exposure time (fast shutter speed). In order to have a short exposure and still see anything, we need to ensure that the subject is well lit and very bright. Consider manually setting the shutter speed for the best result.

Practical part

REQUIREMENTS You and your group will:

- select a (simple) experiment inspired by a mechanical engineering problem or physical event among the examples listed
- build the experimental set-up
- conduct measurements
 - **1 image-processing based measurement**
 - **sensor measurements using 2 different sensors**

The sensors should be picked from different groups listed here:

- load cell
- accelerometer or gyroscope
- (differential) pressure sensor
- hall sensor
- temperature sensor
- (infrared) distance sensor, time of flight sensor
- turbidity, lux, pH, total dissolved solids, or colour sensor

Practical part

REQUIREMENTS You and your group will:

- select a (simple) experiment inspired by a mechanical engineering problem or physical event among the examples listed
- build the experimental set-up
- conduct measurements
 - **1 image-processing based measurement**
 - **sensor measurements using 2 different sensors**

B. Sensor measurements

Conduct two sensor based measurement using two sensors from the following list (or make your own suggestion):

- Load cell: measure the drag force exerted on the strip.
- Differential pressure sensor: measure the dynamic pressure induced by the flow.
 - When a moving parcel of fluid is stopped, its kinetic energy is converted to pressure. By measuring the difference between ambient pressure and the pressure at a point where the flow is stopped (stagnation point), we get a measure of the dynamic pressure induced by the deceleration of the fluid parcels. The dynamic pressure is therefore linked to the flow velocity in the free stream. The sum of ambient (static) pressure and dynamic pressure is known as total pressure.
 - Well aligning the tube inlet with the flow is important for getting an accurate measure of the stagnation pressure.
- Temperature (hairdryer only): measure the temperature of the incoming flow.
- Turbidity (water channel + PIV only): measure the turbidity of the water in the water channel, before and after seeding the flow with tracer particles.

Practical part

REQUIREMENTS You and your group will:

- select a (simple) experiment inspired by a mechanical engineering problem or physical event among the examples listed
 - build the experimental set-up
 - conduct measurements
 - **1 image-processing based measurement**
 - **sensor measurements using 2 different sensors**
 - **an optical field measurement**
- using one of the following techniques:
- **Particle image velocimetry (PIV)** to measure the velocity field in a fluid
 - **Background oriented schlieren (BOS)** to visualise density gradients
 - **Digital image correlation (DIC)** to measure the displacement and strain fields in a deforming solid
 - **Photoelasticity** to visualise stress distributions in birefringent materials

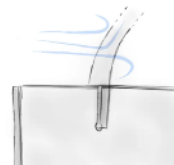
C. Optical field measurement

Example optical field measurements that can be conducted in this type of set-up:

- PIV (water channel only): measure the flow field around or in the wake of the strip.
- BOS (hairdryer only): visualize the density gradients in the hot exhaust of the hairdryer.
- Photoelasticity: visualize the internal stresses within the plate holding the strip, induced by the drag.

To do photoelastic measurements, you need a material with photoelastic properties. The holding plate must e.g. be cut from a single piece of polycarbonate (PC) sheet in a fashion similar to the drawing.

Care must be taken into fastening the strip in a way that does not generate excessively large static stresses in the polycarbonate plate.



Practical part

REQUIREMENTS You and your group will:

- select a (simple) experiment inspired by a mechanical engineering problem or physical event among the examples listed
- build the experimental set-up
- conduct measurements
 - **1 image-processing based measurement**
 - **sensor measurements using 2 different sensors**
 - **an optical field measurement**
- perform an error analysis and characterise the sensitivity, accuracy, etc. for each measurement

Practical part

REQUIREMENTS You and your group will:

- select a (simple) experiment inspired by a mechanical engineering problem or physical event among the examples listed
- build the experimental set-up
- conduct measurements
 - **1 image-processing based measurement**
 - **sensor measurements using 2 different sensors**
 - **an optical field measurement**
- perform an error analysis and characterise the sensitivity, accuracy, etc. for each measurement
- perform a frequency analysis on the data from one of the sensor measurements according to the instructions given in the master class by Dr. Müllhaupt

Practical part

REQUIREMENTS You and your group will:

- select a (simple) experiment inspired by a mechanical engineering problem or physical event among the examples listed
- build the experimental set-up
- conduct measurements
 - **1 image-processing based measurement**
 - **sensor measurements using 2 different sensors**
 - **an optical field measurement**
- perform an error analysis and characterise the sensitivity, accuracy, etc. for each measurement
- perform a frequency analysis on the data from one of the sensor measurements according to the instructions given in the master class by Dr. Müllhaupt
- document your methods and results in a written report

ASSESSMENT A technical report describing the design and characteristics of the measurements systems used and a summary of the measured data will count for 70 % of your final grade.

Practical part

FOCUS The focus should be on the practical design of the experiment, the selection and correct application of the measurement techniques, and the signal conditioning and analysis. **You should not worry about using your data to solve an engineering question or problem** (that is for a different course). Through this course you should gather theoretical and practical knowledge on basic measurement principles and the commonly used engineering measurement systems.

RESOURCES Each group will receive an Arduino kit and will have access to sensors and motors to design their experiment. A summary of the available material can be found on . You can use the resources at SPOT, including 3D printers and laser cutting equipment, according to the SPOT guidelines. Short tutorials to use the most common equipment are also available on .

Practical part

SPOT stands for *Student Prototyping and Outreach Tank*. Find a series of videos on how to access and use the SPOT facilities here .

All of you should follow the following mandatory trainings:

1. General training  (add project make ME301 when following this tutorial)
2. *General introduction to safety and prototyping* on Moodle 

When you have successfully passed these trainings, you will see a **green light** when presenting your CAMIPRO card at the SPOT entrance.

It is recommended that at least **one member per team** follows additional training for *laser cutting* and for *3D printing*. These trainings will be organised by the SPOT coaches during the first weeks of the semester.

Practical part

Manage your time For the practical part of the course, we counted roughly a total estimated workload *per person* in the group of 80 h or a little over 6 h/Friday that we are in session this semester.

We suggest planning approximately:

- 2 sessions for the design and construction of the experiment
- 2 sessions for image-processing based measurements (testing, debugging, data collection, and error analysis)
- 2 sessions for sensor measurements (testing, debugging, data collection, and error analysis)
- 1 session for frequency analysis of the temporal sensor measurement
- 3 sessions for optical field measurement (testing, debugging, data collection, and error analysis)
- 2 sessions for report writing
- 1 session buffer for repeating, improving, catching up, mid-term update,

(= 13 sessions)

Practical part

Manage your time For the practical part of the course, we counted roughly a total estimated workload *per person* in the group of 80 h or a little over 6 h/Friday that we are in session this semester.

We suggest planning approximately:

- 2 sessions for the design and construction of the experiment
- 2 sessions for image-processing based measurements (testing, debugging, data collection, and error analysis)
- 2 sessions for sensor measurements (testing, debugging, data collection, and error analysis)
- 1 session for frequency analysis of the temporal sensor measurement
- 3 sessions for optical field measurement (testing, debugging, data collection, and error analysis)
- 2 sessions for report writing
- 1 session buffer for repeating, improving, catching up, mid-term update,

(= 13 sessions)

Important dates are listed on 

- Halfway through the semester, a mandatory progress meeting will be scheduled with the teaching team to discuss challenges, provide feedback, and evaluate your progress.
Each group will be requested to answer a questionnaire in the week prior to the meeting.
- Each group has a mandatory *Master class on signal processing* given by Dr. Müllhaupt

Expectations

What can you expect from us?

We do our best to

- provide you with a structure for the course
- expose you to different sources of information to help you understand the core teaching points e.g. course notes, course videos, text books, on-line material etc.
- help you develop critical thinking, problem-solving and debugging skills, and engineering best practice through demonstration (in course videos) and interactive discussion (during the Friday sessions at the SPOT)
- provide an environment where the knowledge and techniques from the course can be developed
- test you on the course material and set a standard that you need to pass

Expectations

What do we expect from you?

Experiments typically do not run flawlessly from the first time you run a test. It is an iterative process. You will learn the most by failing first. One of the most important skills of a modern engineer is the ability to debug, i.e. figuring out why a design/experiment/code does not work as planned and fix it. This course will help you further develop that skill. So, ...

- it is perfectly okay, and it can be productive to get things wrong, but
- understand that we will not show you how to solve every problem; we will however provide tools to help you solve problems and opportunities to bounce off ideas
- try not to be stubborn but be persistent - if something does not work, you might need to rethink your design/code/choices, this is typically not a waste of time, but a valuable learning experience
- allocate time for yourself and your team to process the course material, to fail, and to debug
- remember that feedback is a source for improvement and not a personal attack or criticism
- try not to respond to feedback with feedback; instead focus on understanding and internalizing the feedback received, considering its validity and relevance, and taking appropriate actions to address issues raised or implement suggested improvements
- remember that dividing tasks is not the same as sharing the workload and collaborating; use this course to further develop your team-working skills and use the diverse perspectives within your team to improve your individual skills and your collective results

Questions



IT'S IN THE ~~SYLLABUS~~



This message brought to you by every instructor that ever lived.

WWW.PHDCOMICS.COM

For all other questions:

1. Ask the teachers or assistants during the Friday sessions at SPOT
2. Contact us 